X3N-O-THON 2026

TEAM NAME : SYNAPSE_GTB

CATEGORY : AI/ML-powered Web Application

PROJECT TITLE : Vital Shield AI

DOMAIN : Healthcare Track

PROBLEM STATEMENT

User Pain Points

Doctors/Nurses often experience:

- **Alarm fatigue** from too many reactive alerts
- Difficulty identifying **subtle deterioration patterns**
- Lack of **predictive insights** from continuous patient data

Patients often experience:

- Long time in collection of computerized data
- Fraud in case of Malpractice suit
- Keeping up with follow ups

As a result, patient deterioration such as **sepsis, shock, or cardiac arrest** is often detected **too late**, reducing survival chances.

Technical / Operational Challenges

Current systems face several limitations:

1. **Reactive Monitoring**
   Traditional monitors trigger alerts only when vitals exceed predefinedthresholds.
2. **No followup or routine checkup**
   Patients rarely go for the follow ups and Routine checkup
3. **No record of administration**
   There is no record of what and when was the patient given a certain medicine
4. **Doctor word against the family**
   At time of malpractice suit, evidence canbe tampered with.
5. **Memory > Record**
   System is memory reliant rather than electronic evidence.

Why This Problem Matters Now

Healthcare systems are becoming increasingly **data-rich but insight-poor.**

Recent trends driving urgency:

- Rising ICU and emergency department load
- Increasing clinician burnout
- Availability of large clinical datasets like **MIMIC-IV**
- Advances in **machine learning for healthcare**

AI can now detect **patterns invisible to human monitoring**, enabling earlier intervention.

PROPOSED SOLUTION

CORE IDEA / CONCEPT

Project Chronos is an AI-driven early Tamper Proof system, whose goal is to predict diseases even before they occur and provide unaltered evidence...which provides relief to doctors and nurses and justice to mistreated patients.

Instead of reacting to crises, the system **anticipates risk** and alerts clinicians in advance...and instead of word of mouth we have a SHA256 chain run

KEY FEATURES

1. Early Risk Prediction

Machine learning models analyze patient vitals to estimate the probability of deterioration.

2. Continuous Monitoring

The system continuously processes incoming patient data and updates risk scores in real time.

3. Intelligent Alerting

Instead of threshold alarms, the system generates **predictive alerts** based on risk trajectories.

4. Tamper proof surgery records

We provide the hash key value to the patient. In event of the tamper they just have to **validate.**

5. Explainable AI

The system highlights **which vital signs contributed most to the prediction**, improving trust among clinicians.

HOW IT ADDRESSES THE CHALLENGES

Challenge	Solution
Reactive monitoring	Predictive machine learning models
Data overload	Automated risk scoring
Alarm fatigue	Intelligent alerts based on trends
Complex data interpretation	Visual dashboards

WHY THIS SOLUTION IS BETTER

Existing hospital monitoring systems mainly rely on **fixed thresholds.**

Example:

Alert if heart rate > 130

This allows clinicians to **intervene earlier**, potentially preventing life-threatening events.

Predict probability of deterioration in next 6 hours

THE INNOVATION

WHAT MAKES IT UNIQUE / ANTI-BORING

Three aspects make the solution stand out:

1. Predictive Healthcare

Moves from **reactive medicine → predictive medicine**

2. Real Clinical Data

The system is built using the **MIMIC-IV dataset**

3. Scalable Architecture

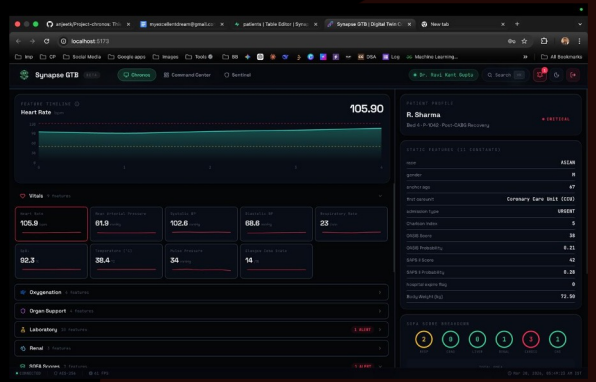
The system is designed to integrate with **hospital electronic health record systems** (77 features)

4.Tamper Proof validation

A little change in the data changes the hash number.

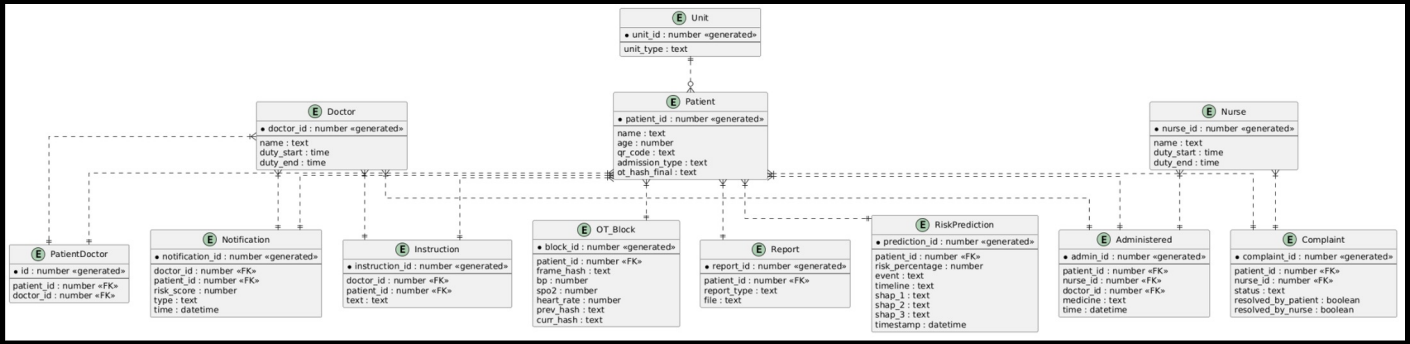
5. Easy boarding, and quick reliability

QR code and scanning easy onboaring...desgined for technically illiterate people.



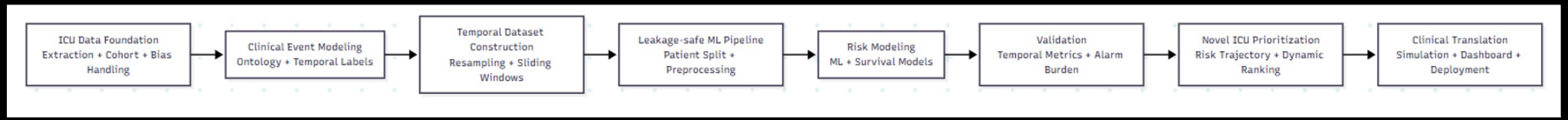
TECHNICAL DESCRIPTION::ERD::ARCHITECTURE

Category	Technology
Prog.Language	Python
Data Processing	Pandas(400 gb database)
Machine Learning	XGBoost
Visualization	Frontend tech, webrtc
Database	BigQuery
Dataset	MIMIC-IV-ED Demo
Deployment	Vercel



Ranking Score Calculation:

ISP = risk_score * weight * time_horizon



IMPACT & BENEFITS

EXPECTED IMPACT

We are redefining how hospital operating systems function by introducing a tamper-proof framework that ensures transparency and accountability. This system aims to build trust with patients' families, streamline patient flow, and reduce the operational burden on nurses, enabling them to deliver care more efficiently and effectively.

SHORT-TERM VS LONG-TERM VALUE

Short Term

- Prototype demonstrating predictive monitoring
- Tamper proof evidence
- Work life balance culture for nurses

Long Term

- Integration into hospital monitoring systems
- Deployment in ICU and emergency departments
- Large-scale predictive healthcare systems

MEASURABLE IMPROVEMENTS

Potential measurable improvements include:

- earlier detection of deterioration
- reduced ICU mortality
- reduced emergency response time
- improved clinician decision support

Benefit	Explanation
Speed	Automated real-time risk predictions
Accuracy	Machine learning detects complex patterns
Simplicity	Easy-to-use clinical dashboard
Accessibility	Scalable cloud-based deployment
Early Intervention	Enables preventive treatment

FEASIBILITY & VIABILITY

TECHNICAL FEASIBILITY

The system is technically feasible due to:

- publicly available clinical datasets
- mature machine learning frameworks
- scalable cloud infrastructure
- No usage of any api key and light weight model.

Auroc : 0.91

Aurpc : 0.83

SCALABILITY & FUTURE ENHANCEMENTS

Future extensions may include:

- training on **full MIMIC-IV dataset**
- deep learning models (LSTM / Transformers)
- real-time hospital data integration
- wearable device monitoring
- personalized patient risk modeling

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